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M. Salih Sayal

github.com/salihsayal
LinkedIn Profile

Experience

IT Engineer Automation — **Everllence (formerly MAN ES)** — **06/2025 – 09/2026**

- **Data Engineering & ETL:** Developed a **Docker-based Python pipeline** to automate complex billing workflows (PostgreSQL) incl. **Active Directory (LDAP)** synchronization.
- **Backend Development:** Designed and implemented a **REST API** using **Java Spring Boot** for seamless integration with **SAP systems**.
- **System Integration:** Developed specialized **wrapper scripts** in Python to orchestrate intercommunication between heterogeneous software systems.
- **Cloud Automation & Reporting:** Operated workloads on **Microsoft Azure** (VMs, Entra ID) and automated data reporting via **Microsoft Graph API** (OAuth2) and Plotly.
- **Disaster Recovery & Scripting:** Implemented automated **backup/restore processes (SFTP/Rsync)** in Linux server environments and refactored **Bash scripts**.

Software Engineer — **rola Security (T-Systems)** — **10/2023 – 04/2025**

- **DevOps & Troubleshooting:** Performed **error analysis** during system startup and automated configuration deployments in Linux environments.
- **Data Engineering:** Developed **Java-based analysis tools** for the automated evaluation of complex system error logs.
- **Software Quality Assurance:** Ensured high quality standards by maintaining **Checkstyle rules** and implementing **Sonar recommendations** (Clean Code).

Projects

Infrastructure & Cloud Project — **Jobentra** — **03/2026 – 09/2026**

- **Kubernetes Cluster & GitOps:** Setup and administration of a stateless **3-node Kubernetes cluster** on Ubuntu Server for microservices, including **GitOps pipelines** via **ArgoCD**, Gitea, and CI/CD runners.
- **Proxmox VE & HA Architecture:** Designed a highly available infrastructure on Proxmox VE with **active replication** and failover on dedicated primary and fallback hardware.
- **Stateful Services & IAM:** Operated core infrastructure services such as **Samba AD DC (SSO)**, **NFS storage** for Kubernetes Persistent Volumes, PostgreSQL, and Nginx Reverse Proxy.
- **Full-Stack CRM Development:** Designed and AI-assisted the implementation of a CRM software using **Java Spring**, **Python FastAPI (Vertex AI / Gemini)**, **PostgreSQL**, and **Next.js**.

Technical Skills

- **Languages & Frameworks:** Java (Spring Boot), Python (**FastAPI**, Pandas, Plotly), SQL, JavaScript / TypeScript, **Next.js**, Bash, PowerShell
- **Infrastructure, Cloud & DevOps:** Kubernetes (K3s/K8s), ArgoCD (GitOps), Proxmox VE, Docker / Compose, **Microsoft Azure**, Linux Administration (Debian/Ubuntu, Arch, RHEL/Fedora), ZFS, CI/CD
- **Security & Networking:** Identity Management (**Active Directory, LDAP, Samba AD DC**), Nginx Reverse Proxy, OPNsense Firewall, WireGuard / Tailscale VPN, NFS Storage, SSH, TLS, DNS
- **Tools & Quality:** Git (Gitea/GitHub/Bitbucket), Jira, Confluence, OpenProject, API Integration (REST, MS Graph, SAP), JUnit 5, Mockito, SonarQube, Checkstyle

Education

B.Sc. Computer Science — **HHU Düsseldorf** — **exp. Spring 2027**

- Current GPA: **2.4** (148 ECTS)
- **Focus Areas:** Software Engineering (Java/C), Databases (SQL/NoSQL)
- **Special Achievements:** Programming (Grade: 1.0), Databases: Advanced Concepts (Grade: 1.0)
- **Electives:** Distributed Systems, Operating Systems, Databases & NoSQL

Voluntary Military Service — **German Armed Forces (Bundeswehr)** — **02/2021 – 06/2021**

General University Entrance Qualification — **Elly-Heuss-Knapp-Gymnasium Duisburg** — **2020**

Languages

- German (Native)
- English (Fluent)
- Turkish (Fluent)
- Japanese (Pre-intermediate)